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Estimating the extent of cascading transmission line outages using standard utility data and a branching process

Ian Dobson, *Fellow IEEE*
Electrical and Computer Engineering Department
University of Wisconsin, Madison WI 53706 USA
dobson@engr.wisc.edu

***Abstract*—The number of transmission line outaged is one measure of the extent of a cascading disturbance. An initial number of lines outaged can lead to a cascading propagation of further outages. We estimate the amount by which line outages propagate from standard utility data that is reported annually to NERC by transmission owners for the Transmission Availability Data System. Given an assumption about the initial lines outages, we estimate the distribution of the total number of outages after cascading from the amount of propagation and a probabilistic branching process model of the cascading.**

***Index Terms*—power system monitoring, power transmission reliability, failure analysis, risk analysis.**

This paper gives an overview for a panel session. Some more detailed explanations are given in an explanatory way in the Appendix and in more mathematical detail in [1].

Transmission line outages are useful diagnostics in monitoring the progress and extent of blackouts. One common feature of large blackouts is the successive failure of transmission lines, and the number of transmission lines outaged is a measure of the blackout extent. Also, the transmission line outages that do not lead to load shed can be regarded as precursor data for the transmission line outages that do lead to load shed and blackout.

Transmission owners in the USA are required to report transmission line outage data to NERC for the TADS Transmission Availability Data System. The TADS data for each transmission line outage includes the outage time to the nearest minute. The example of transmission line outage data used in this paper is 8864 outages in TADS data recorded by a North American utility over a period of ten years [1], [2]. All the line outages are automatic trips and most of the outages are of lines rated 115 kV or above.

The analysis supposes that the cascading line outages occur in generations following some initial line outages. That is, the initial line outages are generation zero, and then cascading produces further line outages in generation one, and then in generation two, and so on. The utility line outage data is processed in a simple way by separating the line outages into distinct cascades and then separating each cascade into generations or stages. This is done according to the timing of the line outages; line outages in quick succession are in the same generation of a cascade, and a succession of line outages with no hour-long gaps are in the same cascade. It is then easy to obtain statistics of the number of line outages in the generations as shown in Table I.

TABLE I
NUMBER OF LINE OUTAGES IN GENERATIONS 0 TO 10 SUMMED OVER ALL THE CASCADES

generation number	0	1	2	3	4	5	6	7	8	9	10
number of outages	6254	1143	434	227	155	95	78	53	46	32	31

The average propagation at each generation is easily obtained from Table I. For example, propagation from generation zero to generation one is defined to be the average number of line outages in generation one per outage in generation zero. In the utility data set, there are 6254 outages in generation zero of a cascade and 1143 outages in generation one of a cascade. Therefore the generation one propagation is estimated to be 1143/6254 = 0.18. There are 1143 outages in generation one of a cascade and 434 outages in generation two of a cascade. Therefore the generation two propagation is estimated to be 434/1143 = 0.38. The propagation for each generation is calculated in a similar way for the subsequent generations [1] and is shown in Table II.

TABLE II
PROPAGATION IN EACH GENERATION ESTIMATED FROM UTILITY DATA

generation number	1	2	3	4	5 or more
propagation	0.18	0.38	0.52	0.68	0.75

In Table II, the amount of propagation varies with generation number as the cascade proceeds, increasing starting from 0.18 at generation one to about 0.75 at generation five and subsequent generations. This increase in propagation as the cascade proceeds is a significant observation. It would be worthwhile to find out if this observation is typical for other utility data sets, if these could be made available for analysis.

Branching process models are bulk statistical models of the cascading process. Branching process models for cascading line outages are analytically and computationally tractable. Computer algebra can be used to estimate the distribution of the total number of line outages from assumed initial outages and estimates of the propagation [1]. Branching process models are largely consistent with observed and simulated data for the purpose of estimating the distribution of the final blackout size [1], [3], [4]. In particular, given the propagation estimated for each generation and the initial distribution of line outages from the utility data, a branching process can estimate the distribution of the total number of line outages, and this

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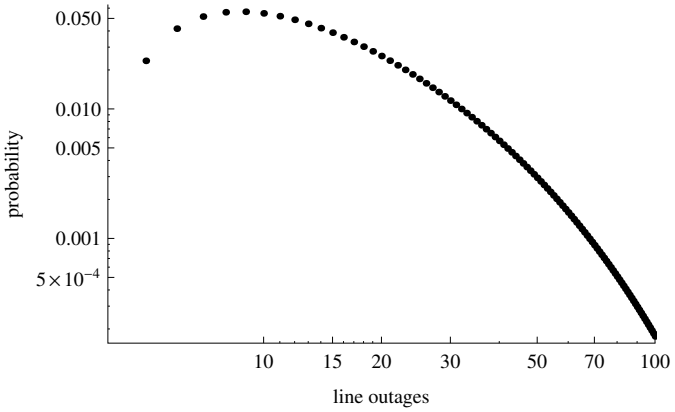


Fig. 1. Probability distribution of the total number line outages assuming five initial line outages. Result obtained using a branching process model with the propagation estimated from the utility data set.

distribution is consistent with the empirical distribution of the total number of line outages [1]. That is, for the utility data set studied, a branching process model that accounts for the varying propagation as the cascade progresses can give a good prediction of the distribution of the total number of line outages from the initial number of line outages. Branching processes have also been used historically to model cascading processes in many subjects outside risk analysis [5], but their application to cascading failure is recent [6].

A significant outcome of the estimation of propagation from the utility data is the ability to estimate the consequence after cascading of some assumed initial line outages. For example, suppose there are five initial line outages and the propagation is estimated from the utility data set. Then the branching process model can be used to estimate the distribution of the total number of line outages as shown in Figure 1. The calculation is indicated in the Appendix and details are given in [1]. Instead of assuming some initial line outages, conventional risk analysis and observational methods could be used to give the distribution of the initial number of line outages.

Cascading failure can be thought of as initial outages that then propagate in generations. Many of the initial outages are due to component reliability or external stresses such as weather, whereas the propagation of outages is more related to the overall resilience of the system. To mitigate cascading failure, it is necessary either to limit the initial outages or reduce the propagation. One benefit of quantifying both the propagation in generations and the resulting distribution of total line outages is that the relative effectiveness of mitigating propagation at different generations can be evaluated. For the utility data set, it appears that to mitigate long cascades it is more effective to reduce the amount of propagation at the early stages of cascading [1]. It would also be useful to compare the effect of limiting the initial outages with the effect of limiting propagation. Little is known about whether propagation can effectively be changed to improve cascading performance, but the quantification of propagation and its impact on blackout extent discussed here is a first step towards this.

Why not simply estimate the distribution of the total number of line outages empirically? The reason is that accumulating enough data to get statistically accurate estimates takes too long, especially for the larger cascades which are rarer. Analysis of the utility data set suggests that it requires an order of magnitude more data to estimate the distribution of total number of line outages empirically than estimating the propagation and using the branching process. That is, to estimate, say, the probability of a total of 10 line outages to a given accuracy with the branching process requires less than one tenth of the data required for empirical estimation. The number of line outages that have to be observed to get sufficiently accurate estimates of the distribution of the total number of line outages is a key quantity that determines the practicality of monitoring the power system with these methods.

The statistical accuracy of the estimated distribution of the total number of line outages can be increased either by gathering line outage data over a larger region or observing the line outage data for a longer time. Of course the resulting estimation applies as a bulk analysis averaged over both the chosen region and the chosen time span. Note that although the distribution of the total number of line outages is estimated, there is no information about which line outages will occur. This implies that there is no conclusion directly from the analysis itself of how to best to improve the cascading performance of the power system, although there is clearly scope for studying this relationship and learning how to do this. If the initial failures are not assumed, the statistical accuracy of the total number of failures is also affected by any uncertainty in the estimates of the initial failures. The goal is to estimate the distribution of total number of line outages with useful accuracy from one year of data. Initial results based on the utility data set suggest that this goal can be achieved for the Western interconnection of North America and some large subregions of the Western interconnection.

In summary, based on an observed utility data set and branching process model, we demonstrate a new capability to quantify the propagation of cascading line outages from standard utility data and estimate the distribution of the total number of line outages due to cascading from known or assumed initial line outages.

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APPENDIX: SUMMARY OF METHOD

A. Branching process modeling and calculations

This appendix informally explains the branching process model and calculations that are described more precisely in [1]. This appendix tries to limit the required background in branching processes, but some basic knowledge of branching processes is helpful. Chapter I of [5] is the classic explanation.

As applied in this paper, the input parameters of the branching process are the distribution of the initial outages in generation zero and the average propagation λ_k in generation number k of the branching process. The goal is to estimate the distribution of the total number of line outages.

The branching process model is a Galton-Watson branching process with discrete time marking the generations and a discrete number of line outages in each generation. The Galton-Watson branching process is a Markov chain and the main interest is in the transient cascading process and its outcome. The offspring distribution of the branching process is the probability distribution of the number of line outages that are produced in the next generation by one line outage. Each line outage in each generation produces its share of the next generation according to independent samples of the offspring distribution. These assumptions are used in the branching process, but there is no intent to model the details of the cascading or represent notions of one line outage "causing" another line outage with these assumptions. The way in which the branching process is proposed to model the cascade is simply that it can reproduce the statistics of the number of line outages in each generation and the statistics of the total number of line outages.

The offspring distribution for generation number k is a Poisson distribution with mean λ_k. The Poisson distribution is chosen because it is a good model when there are many similar and discrete outcomes, each of low probability. That is, each line outage could produce many other line outages, each of low probability. The mean offspring or average propagation λ_k varies with the generation as the cascade proceeds because that is what we observe in the data.

The Poisson offspring distribution for generation k with mean λ_k is given by the following power series in s, which is called a generating function:

$$e^{-\lambda_k+\lambda_k s} = e^{-\lambda_k} + e^{-\lambda_k}\lambda_k s + e^{-\lambda_k}\frac{\lambda_k^2 s^2}{2!} + e^{-\lambda_k}\frac{\lambda_k^3 s^3}{3!} + ...$$

The coefficient of s^r in the generating function is the probability of r line outages being produced.

The calculations for branching processes are elegant in form, and involve the nesting or functional composition of the generating functions. For example, suppose that there is only one line outage in generation zero. Then the probability distribution of the total number of line outages produced by four generations of cascading has a generating function given by the formula

$$se^{-\lambda_1+\lambda_1 se^{-\lambda_2+\lambda_2 se^{-\lambda_3+\lambda_3 se^{-\lambda_4+\lambda_4 s}}}} \quad (1)$$

The nesting of the generating functions in (1) mirrors the stages of the cascade. Expanding (1) as a power series in s gives the coefficients of power of s which are the probabilities of the total number of line outages. For example, the coefficient of s^{12} in the power series expansion of (1) is the probability of a total of 12 lines outaged. In practice, this calculation is done using computer algebra.

The calculation of the probability distribution of the total number of outages used in the paper is similar to (1), except that the calculation needs to account for an infinite number of generations with $\lambda_k = 0.75$ for $k \geq 5$. This is done using the fact that a Poissonian Galton-Watson branching process with constant propagation λ_k has total number of line outages given by the Borel distribution with parameter λ_k.

B. Branching process validation and application

The branching process model is *validated* on the longest available data set, in this case the full decade of line outage data. The propagations and the probability distribution of the initial line outages are estimated from the data, and the branching process model is used to estimate the probability distribution of the total number of line outages. For comparison, the full data is again used to empirically estimate the probability distribution of the total number of line outages. If the empirical and branching process probability distributions match well, as they do for the data set of this paper [1], this validates this use of the branching process model. Of course, it would be very desirable to perform the validation on many data sets, but data confidentiality concerns are a limiting factor at present. This paper helps to explain the method so that the branching process model can be more widely tested in the future.

To *apply* the branching process model, the results on the data set used in the paper suggest that it is practical to estimate the propagations from a shorter data record of about one year.

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